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Math 102 Final Exam Term 223

Original exam date: 17 August 2023. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. If $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-5)^n}{n\,5^n}$, then the interval of convergence for $f(x)$ is

(a) $(-5, 5]$

(b) $[0, 10)$

(c) $(0, 10]$

(d) $(-5, 5)$

(e) $(0, 10)$

2. If $G(x) = \int_0^x \left[s \int_0^s f(t)\,dt\right] ds + \int_0^{2x} t\,dt$, then $G''(0) =$

(a) 1

(b) 0

(c) $G'(0)$

(d) 4

(e) 2

3. Which one of the following statements is True?

(a) It is possible to find a power series whose interval of convergence is $[0, \infty)$.

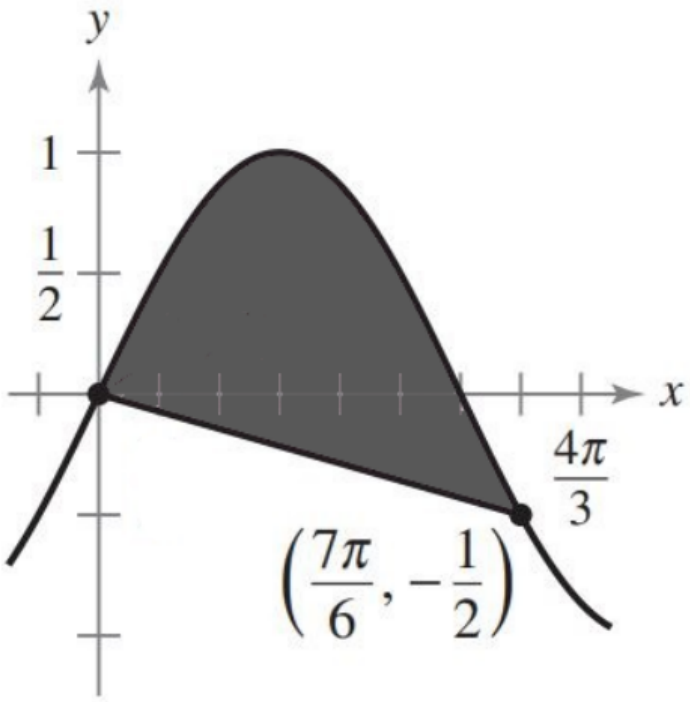
(b) The radius of convergence for $\sum_{n=1}^{\infty} \frac{1}{n}$ is 1.

(c) If the power series $\sum_{n=1}^{\infty} a_n x^n$ converges for $x = 2$, then it also converges for $x = -2$.

(d) If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges for $|x| < 2$, then $\int_0^1 f(x)\,dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1}$.

(e) If the interval of convergence for $\sum_{n=0}^{\infty} a_n x^n$ is $(-1, 1)$, then the interval of convergence for $\sum_{n=0}^{\infty} a_n (x-1)^n$ is $(-2, 0)$.

4. If the area between the graph of $y = \sin x$ and the line segment joining the point $(0, 0)$ and $\left(\frac{7\pi}{6}, -\frac{1}{2}\right)$ (shown in the figure) equals $\frac{\sqrt{3}}{2} + \frac{7\pi}{24} + k$, then $k =$



(a) 1

(b) 4

(c) 2

(d) 3

(e) 5

5. If the region bounded by $y = x^2$, $x = 0$, and $y = 9$ in the 1st quadrant is revolved about the line $x = 3$, then the volume of the resulting solid is

(a) $\frac{27\pi}{4}$

(b) $\frac{113\pi}{5}$

(c) $\frac{135\pi}{2}$

(d) $\frac{143\pi}{6}$

(e) $\frac{123\pi}{3}$

6. The area of the surface generated by revolving the graph of $y = 3x$ over $[0, 3]$ about the x-axis is

(a) $9\sqrt{10}\,\pi$

(b) $3\sqrt{10}\,\pi$

(c) $27\sqrt{10}\,\pi$

(d) $18\sqrt{10}\,\pi$

(e) $6\sqrt{10}\,\pi$

7. $\int_0^1 \frac{dx}{2\sqrt{3-x}\,\sqrt{x+1}} =$

(a) $\dfrac{\pi}{6}$

(b) $\dfrac{\pi}{3}$

(c) $\dfrac{\pi}{2}$

(d) $\dfrac{\pi}{4}$

(e) $\dfrac{\pi}{12}$

8. $\dfrac{1}{10}\sum_{n=1}^{\infty} n\left(\dfrac{9}{10}\right)^n =$

(a) $\dfrac{1}{10}$

(b) 9

(c) 90

(d) 10

(e) $\dfrac{9}{10}$

9. Which one of the following is False about the sequence with the nth term $a_n = \dfrac{\sin\sqrt{n}}{\sqrt{n}}$?

(a) not monotonic

(b) a_n approaches 0 as $n \to \infty$

(c) converges

(d) diverges

(e) bounded

10. Applying the ratio test to the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\left(\frac{\pi}{2}\right)^n}{n^2}$, we get $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$

(a) $= 0$

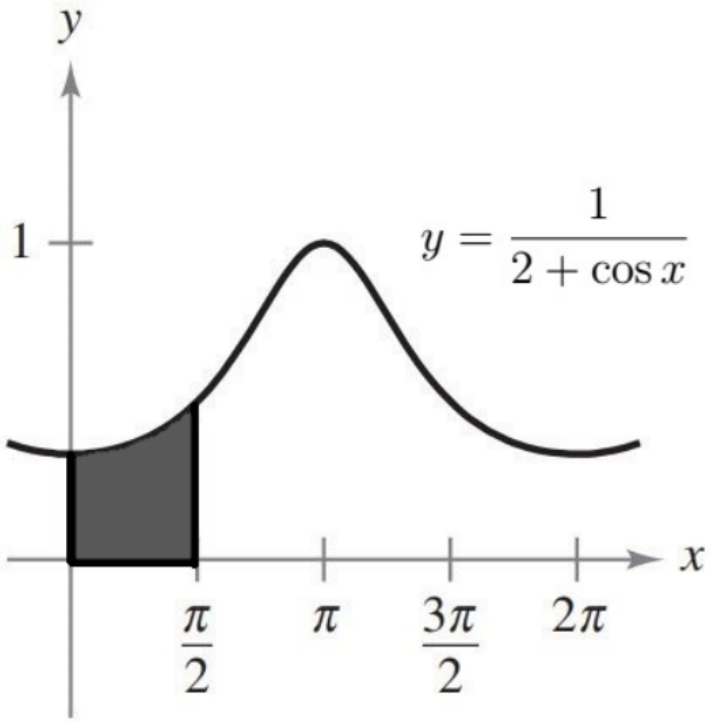
(b) $= 1$ and then the ratio test is inconclusive

(c) > 1 and then the series diverges

(d) $= \infty$

(e) < 1 and then the series converges

11. The area of the region bounded by the graph of $y = \dfrac{1}{2 + \cos x}$ (as shown in the figure) and $y = 0$ for $0 \le x \le \dfrac{\pi}{2}$ is



(a) $3\sqrt{3}\pi$

(b) $\dfrac{\sqrt{3}}{2}\pi$

(c) $\dfrac{\sqrt{3}}{9}\pi$

(d) $\dfrac{\sqrt{3}}{27}\pi$

(e) $\dfrac{\pi}{\sqrt{3}}$

12. $\lim_{n\to\infty} \sum_{i=1}^{n} \left[\sqrt{\frac{i^2}{n^2}} \left(\frac{2i-1}{n^2} \right) \right] =$

(a) $\frac{2}{3}$

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{5}{2}$

(e) 2

13. $\sum_{n=0}^{\infty} \left[\left(\frac{3}{4} \right)^n - \frac{1}{(n+1)(n+2)} \right] =$

(a) 4

(b) 3

(c) 5

(d) 2

(e) 1

14. How many of these series

$$\text{(i)} \sum_{n=1}^{\infty} \frac{\ln n}{n+1}, \quad \text{(ii)} \sum_{n=0}^{\infty} \frac{1}{n!}, \quad \text{(iii)} \sum_{n=1}^{\infty} \frac{\sin^2 n}{n^3}, \quad \text{(iv)} \sum_{n=1}^{\infty} \frac{1}{4\sqrt[3]{n} - 1}$$

is (are) divergent series?

(a) 4

(b) 3

(c) 1

(d) 2

(e) 0

15. The volume of the solid resulting from revolving the region bounded by $y = \frac{10}{x^2}$, $y = 0$, $x = 1$ and $x = 5$ about the line $y = 10$ is given by

(a) $\pi \int_1^5 \left(100 - \frac{1}{x^4} \right) dx$

(b) $100\pi \int_1^5 \left(\frac{2}{x^2} - \frac{1}{x^4} \right) dx$

(c) $200\pi \int_1^5 \left(\frac{1}{x} - \frac{1}{x^3} \right) dx$

(d) $2\pi \int_1^5 \frac{100}{x} - \frac{2}{x^4}\, dx$

(e) $10\pi \int_1^5 \left(\frac{1}{x} - \frac{1}{x^3} \right) dx$

16. Which of the following series converges conditionally?

(a) $\sum_{n=0}^{\infty} \left(\frac{\cos n\pi}{n+1}\right)$

(b) $\sum_{n=1}^{\infty} \frac{\cos\left(n\frac{\pi}{3}\right)}{n^2}$

(c) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n}{e^{n^2}}$

(e) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\sqrt{n}}$

17. $\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$

(a) converges to $\ln 2 - \ln 3$

(b) converges to $\ln 3 - \ln 2$

(c) converges to $-\ln 2$

(d) converges to $\ln 2$

(e) diverges

18. If $\int_3^4 x^3\sqrt{x^2-9}\,dx = k\left[\frac{\tan^3\theta}{3} + \frac{\tan^5\theta}{5}\right]_0^{\sec^{-1}\left(\frac{4}{3}\right)}$, then $k =$

(a) 175

(b) 243

(c) 97

(d) 39

(e) 139

19. $\int_3^2 (3-x)\,7^{(3-x)^2}\,dx =$

(a) $\dfrac{-3}{\ln 7}$

(b) $\dfrac{2}{\ln 7}$

(c) $\dfrac{4}{\ln 7}$

(d) $\dfrac{-6}{\ln 7}$

(e) $\dfrac{1}{\ln 7}$

20. The average value of $f(x) = \dfrac{\ln x^2}{x}$ over $[1, e]$ is

(a) $\dfrac{1}{e-1}$

(b) e

(c) $\dfrac{1}{e}$

(d) 1

(e) $\dfrac{1}{e^2}$

21. Which one of the following statements is False when the integral test is applied to $\sum_{n=1}^{\infty} \frac{\arctan n}{n^2+1}$?

(a) The series converges

(b) $f(x) = \dfrac{\arctan x}{x^2+1}$ is continuous for $x \geq 1$

(c) $f(x) = \dfrac{\arctan x}{x^2+1}$ is decreasing for $x \geq 1$

(d) $f(x) = \dfrac{\arctan x}{x^2+1}$ is positive for $x \geq 1$

(e) The series diverges

22. If $P_3(x)$ is the third Maclaurin polynomial for $f(x) = x^2 e^x$, then $P_3(1) =$

(a) 4

(b) 3

(c) 2

(d) 1

(e) 5

23. $\int \frac{x-8}{x^2-x-6}\, dx =$

(a) $\ln\left|\dfrac{(x+2)^2}{x-3}\right| + C$

(b) $\ln\left|\dfrac{x-3}{x+2}\right| + C$

(c) $\ln\left|\dfrac{x+2}{x-3}\right| + C$

(d) $\ln\left|\dfrac{(x-3)^2}{x+2}\right| + C$

(e) $\ln\left|\dfrac{x+2}{(x-3)^2}\right| + C$

24. The coefficient of x^5 in the power series of $f(x) = (\cos x)\ln(1+x)$ is

(a) $\dfrac{3}{40}$

(b) $\dfrac{1}{5}$

(c) $\dfrac{3}{8}$

(d) $\dfrac{5}{24}$

(e) $\dfrac{-1}{6}$

25. If $a_1 = 1$ and $a_{n+1} = \dfrac{\sin n + 1}{\sqrt{n}}\, a_n$, then $\sum_{n=1}^{\infty} a_n$

(a) diverges by the integral test

(b) converges by the integral test

(c) the ratio test is inconclusive

(d) converges by the ratio test

(e) diverges by the ratio test

26. $\int_0^1 e^{-x^3}\, dx =$

(a) $\sum_{n=0}^{\infty} \frac{1}{(3n+1)(2n+1)}$

(b) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(3n+1)\cdot n!}$

(c) $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3n)(2n+1)!}$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(3n+1)(2n+1)!}$

(e) $\sum_{n=1}^{\infty} \frac{1}{(3n+1)^n}$

27. $\sum_{n=1}^{\infty} \frac{x^{2n}}{2n}$

(a) diverges for all $x \in \mathbb{R}$

(b) converges for $-1 < x < 1$

(c) converges for $-1 \leq x < 1$

(d) converges for $x = 1$ or $x = -1$

(e) converges for $x > 1$ or $x < -1$

28. $\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x}\, dx =$

(a) π

(b) $\dfrac{\pi}{4}$

(c) $\dfrac{\pi}{3}$

(d) $\dfrac{\pi}{2}$

(e) $\left(\dfrac{\pi}{2}\right)^2$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Radius 5 about 5; $x=0$ gives $-\sum \frac{1}{n}$ (diverges), $x=10$ alternating harmonic (converges).
2	**(d)**	$G'(x) = x\int_0^x f + 4x$, $G''(x) = \int_0^x f + xf(x) + 4$, so $G''(0) = 4$.
3	**(d)**	Termwise integration is valid since $[0,1]$ is inside the interval of convergence; the others are false.
4	**(a)**	$A = \int_0^{7\pi/6}(\sin x + \frac{3x}{7\pi})dx = \frac{\sqrt{3}}{2} + \frac{7\pi}{24} + 1$, so $k=1$.
5	**(c)**	Shells: $V = 2\pi\int_0^3 (3-x)(9-x^2)\,dx = \frac{135\pi}{2}$.
6	**(c)**	$S = 2\pi\int_0^3 3x\sqrt{10}\,dx = 27\sqrt{10}\,\pi$.
7	**(e)**	$(3-x)(x+1) = 4-(x-1)^2$: $\frac{1}{2}[\arcsin\frac{x-1}{2}]_0^1 = \frac{1}{2}\cdot\frac{\pi}{6} = \frac{\pi}{12}$.
8	**(b)**	$\sum nx^n = \frac{x}{(1-x)^2}$; at $x=0.9$ it is 90, so the value is 9.
9	**(d)**	Squeeze: $\lvert a_n\rvert \le \frac{1}{\sqrt{n}} \to 0$, so a_n converges; the false claim is divergence.
10	**(c)**	Ratio $\to \frac{\pi}{2} > 1$, so the series diverges.
11	**(c)**	$u = \tan\frac{x}{2}$: $\int_0^1 \frac{2\,du}{u^2+3} = \frac{2}{\sqrt{3}}\cdot\frac{\pi}{6} = \frac{\sqrt{3}}{9}\pi$.
12	**(a)**	Sum $= \frac{2}{n^3}\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2n^3} \to \frac{2}{3}$.
13	**(b)**	Geometric sum 4, telescoping sum 1: $4-1=3$.
14	**(d)**	(i) and (iv) diverge (comparison with $\frac{1}{n}$ and $n^{-1/3}$); (ii), (iii) converge: 2.

Q	Ans	Brief explanation
15	**(b)**	Washers: $R = 10$, $r = 10 - \frac{10}{x^2}$, so $V = 100\pi \int_1^5 \left(\frac{2}{x^2} - \frac{1}{x^4}\right) dx$.
16	**(a)**	$\cos n\pi = (-1)^n$: $\sum \frac{(-1)^n}{n+1}$ converges, but $\sum \frac{1}{n+1}$ diverges.
17	**(c)**	Term $= b_n - b_{n+1}$ with $b_n = \ln \frac{n-1}{n}$: sum $= b_2 = -\ln 2$.
18	**(b)**	$x = 3\sec\theta$: $243 \int \sec^4\theta \tan^2\theta \, d\theta = 243 \left[\frac{\tan^3\theta}{3} + \frac{\tan^5\theta}{5}\right]$.
19	**(a)**	$u = (3-x)^2$: $-\frac{1}{2}\int_0^1 7^u \, du = -\frac{1}{2} \cdot \frac{6}{\ln 7} = \frac{-3}{\ln 7}$.
20	**(a)**	$\frac{1}{e-1}\int_1^e \frac{2\ln x}{x} dx = \frac{1}{e-1}\int_0^1 2u \, du = \frac{1}{e-1}$.
21	**(e)**	$\int_1^\infty \frac{\arctan x}{x^2+1} dx = \left[\frac{(\arctan x)^2}{2}\right]_1^\infty$ is finite, so the series converges.
22	**(c)**	$x^2 e^x = x^2 + x^3 + \cdots$, so $P_3(x) = x^2 + x^3$ and $P_3(1) = 2$.
23	**(a)**	$\frac{x-8}{(x-3)(x+2)} = \frac{-1}{x-3} + \frac{2}{x+2}$: $\ln\left\lvert\frac{(x+2)^2}{x-3}\right\rvert + C$.
24	**(a)**	$\frac{1}{5} - \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{24} = \frac{3}{40}$.
25	**(d)**	$0 \le \frac{a_{n+1}}{a_n} \le \frac{2}{\sqrt{n}} \to 0 < 1$: converges by the ratio test.
26	**(b)**	$e^{-x^3} = \sum \frac{(-1)^n x^{3n}}{n!}$; integrating gives $\sum \frac{(-1)^n}{(3n+1)n!}$.
27	**(b)**	Ratio test gives $x^2 < 1$; at $x = \pm 1$ it is $\frac{1}{2}\sum \frac{1}{n}$ (diverges): $-1 < x < 1$.
28	**(b)**	$u = \sin x$: $\int_0^1 \frac{du}{1+u^2} = \arctan 1 = \frac{\pi}{4}$.